Largen Animal Surgery and Practice

Abdominal Surgery

Abdominal Surgery

- Rumenotomy
- Rumen fistulation
- Left and right displacement of abomasum
- Cecal dilation and dislocation
- Intestinal intussusception
- Umbilical hernia and abscess
- Vagal indigestion
- Abdominocentesis
- Liver biopsy
- Anal and rectal atresia
- Rectal prolapse

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Flank Laparotomy

Indications of Flank Laparotomy

- Left-flank laparotomy:
  - Rumenotomy
  - Rumen fistulation
- Right-flank laparotomy:
  - Omentopexy
  - Pyloro-omentopexy
  - Surgical correction of small intestine, cecal and colon
  - Cesarean section
  - Exploratory

Preoperative preparation & anesthesia

- The cow was restrained in a stanchion in a standing position, with its tail tied to the hind leg on the opposite side of the operation.



Preoperative preparation & anesthesia

- Ceftiofur (1.1–2.2 mg/kg), IM
  Flunixin (1.1–2.2 mg/kg), IM
- IV infusion if needed
  (glucose, Ca, saline, methionine)



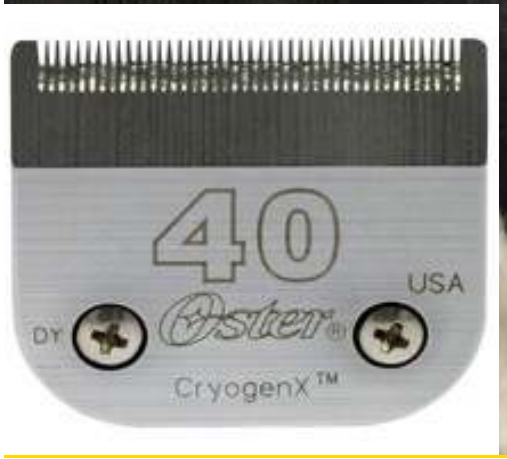


Preoperative preparation & anesthesia

- Shave the hair in the surgical area (15–20 cm from the surgical site, between the 12th intercostal space and the wing of ilium, below the transverse process of the vertebra and the bottom of the flank).
  - Shave off the coarser hair with no. 10 blade.
  - shave off the finer hair with no. 40 blade.


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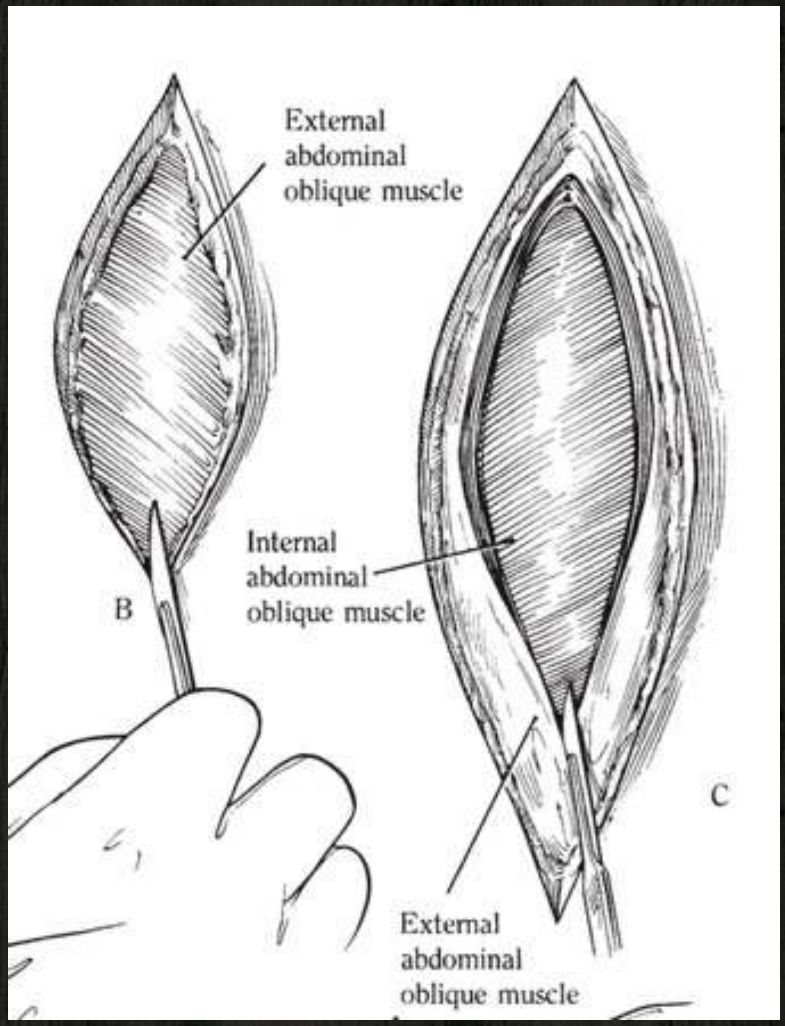
- Linear infiltration anesthesia was performed using 100 mL of 2% lidocaine, covering one subcutaneous layer and two muscle layers.



- Apply povidone-iodine (1% available iodine) solution.



- A vertical incision is made.
  - 3–5 cm (at least) ventral to the transverse processes of lumbar vertebrae
  - 5 cm (at least) caudally to the last rib
  - complete penetration, in one smooth motion.



- Separate the skin and subcutaneous tissue, reveal fibers of the external oblique.

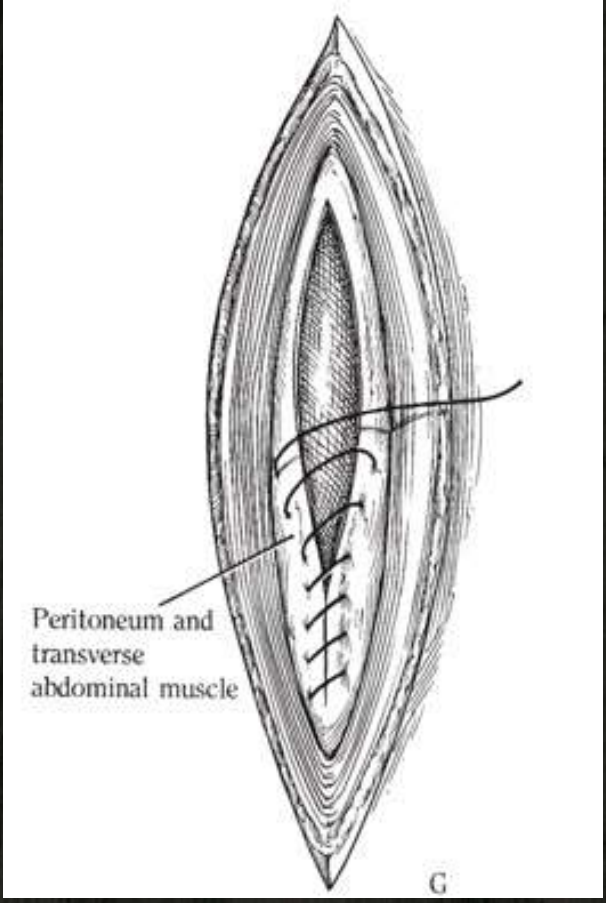


- The external oblique is incised vertically to reveal the internal oblique.



- The internal oblique is bluntly separated according to its muscle direction.
- The transverse abdominis, together with peritoneum, were incised vertically.

Closure



- The peritoneum and transverse abdominis are closed together with a simple continuous suture pattern using synthetic absorbable suture.



- The internal oblique and the external oblique are closed separately with a simple continuous suture pattern using synthetic absorbable suture.



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- Skin closure is performed with interrupted horizontal mattress suture.

Postoperative Management



- The wound is applied with iodine ointment, twice per day, for 3 days.
- Antibiotics are administered, if indicated, depending on the procedure.
- Sutures may be removed 2–3 weeks after surgery.

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Rumenotomy & Rumen fistulation

Purpose of rumenotomy

- To remove foreign bodies, including:
  - rumen or reticulum foreign bodies
  - obstructions between the reticulum and omasum
  - distal esophageal foreign bodies

Surgical Technique

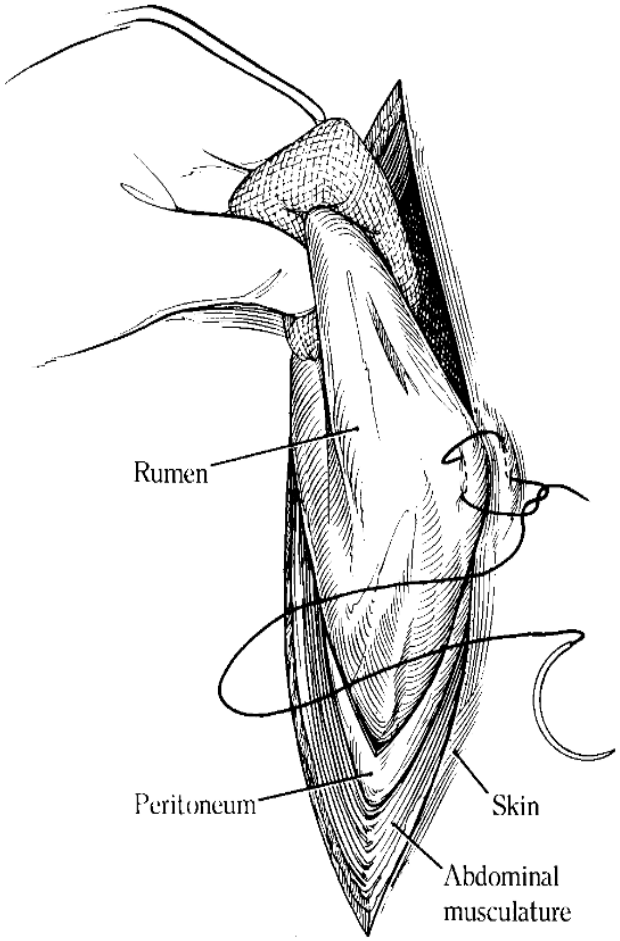
1. Surgical preparation and anesthesia.
2. Open the abdominal cavity according to the left-flank laparotomy procedures.
3. Anchor the rumen to the incision with Cushing pattern.
   - avoid contamination of the muscles and peritoneum during opening the rumen.

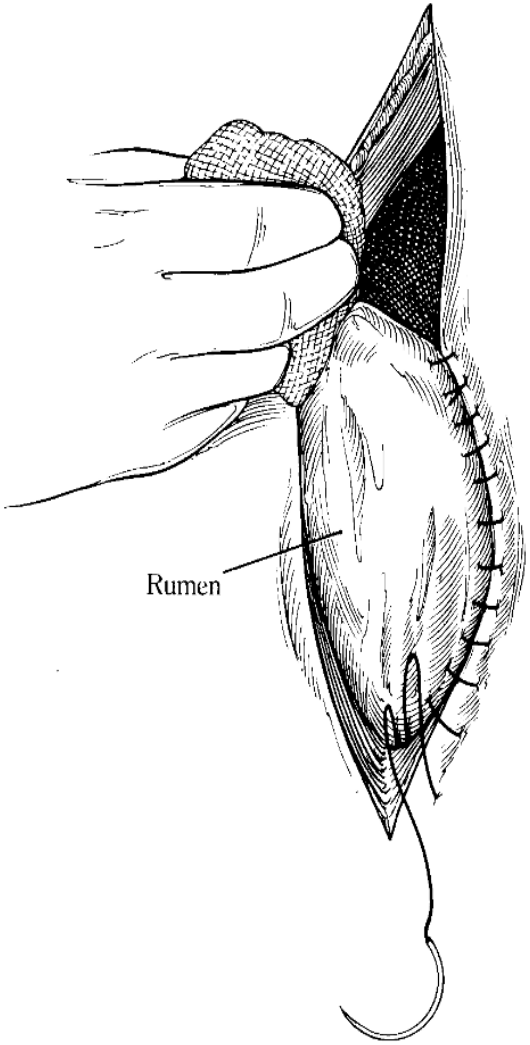
Surgical Technique

4. The rumen is incised with a scalpel.
   - leave enough room dorsally and ventrally for closure.
   - Instruments that come into contact with the rumen mucosa or rumen fluid must be considered contaminated and stored separately from other sterile instruments.
5. The operator, wearing long gloves, evacuates and explores the rumen.
6. Explored the inside of the rumen and the reticulum and if a foreign body is present, it is removed.

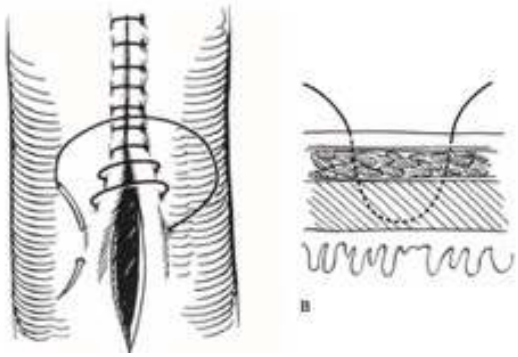
Surgical Technique

4. The rumen incision is closed with a double row:
   - first row: a simple continuous pattern using a no. 1 or 2 synthetic absorbable material.
   - second row: an inverting pattern with similar suture material.
5. The surgical site is then irrigated with sterile saline prior to removal of the rumen-fixation suture.
6. Closure of the laparotomy incision.







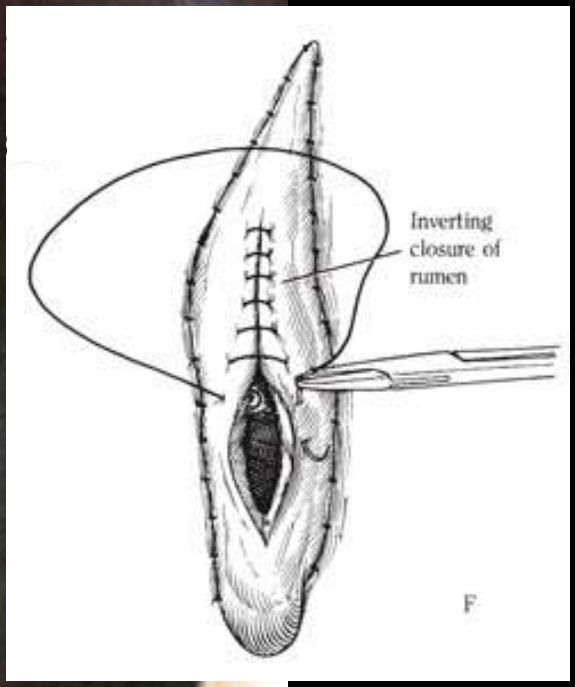


(Cushing pattern)








Inverting
closure of
rumen
F

Purpose of rumen fistulation

- Nutritional trials
- Treatment of chronic bloat
- Transplantation of rumen fluid to the sick cows



Surgical Technique

1. Surgical preparation and anesthesia.
   - administer a sedative if needed
2. Open the abdominal cavity according to the left-flank laparotomy procedures.
3. Anchor the rumen: the serosa of the rumen was sutured and fixed to the skin at the 3, 6, 9, and 12 o'clock positions using No. 3 non-absorbable suture.

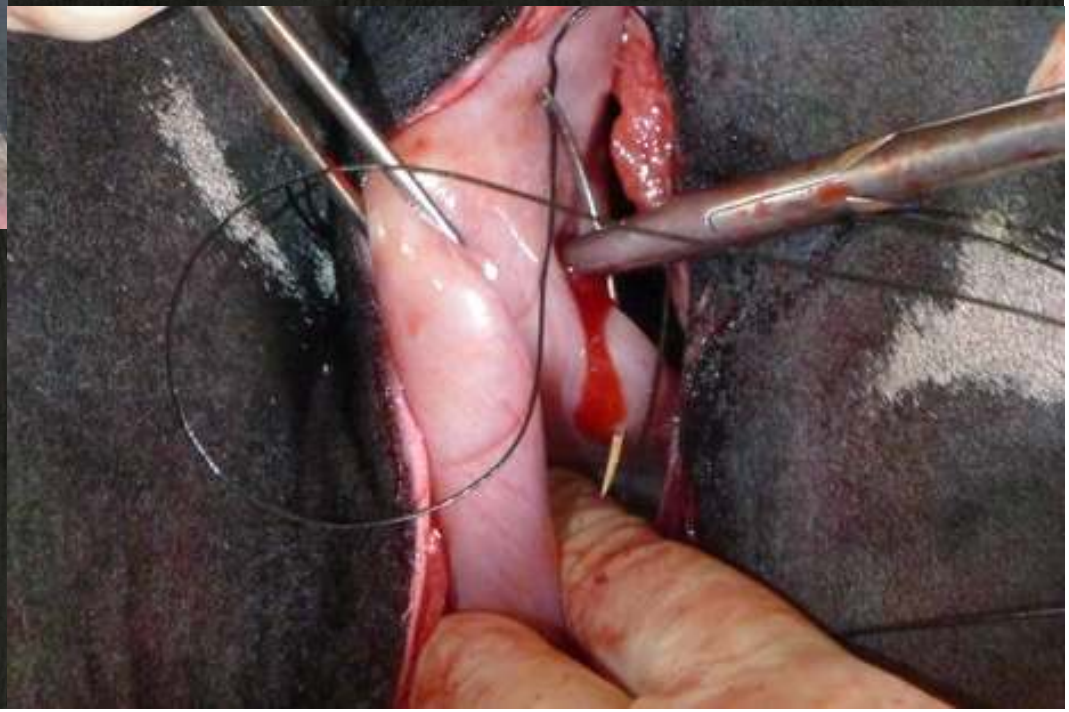










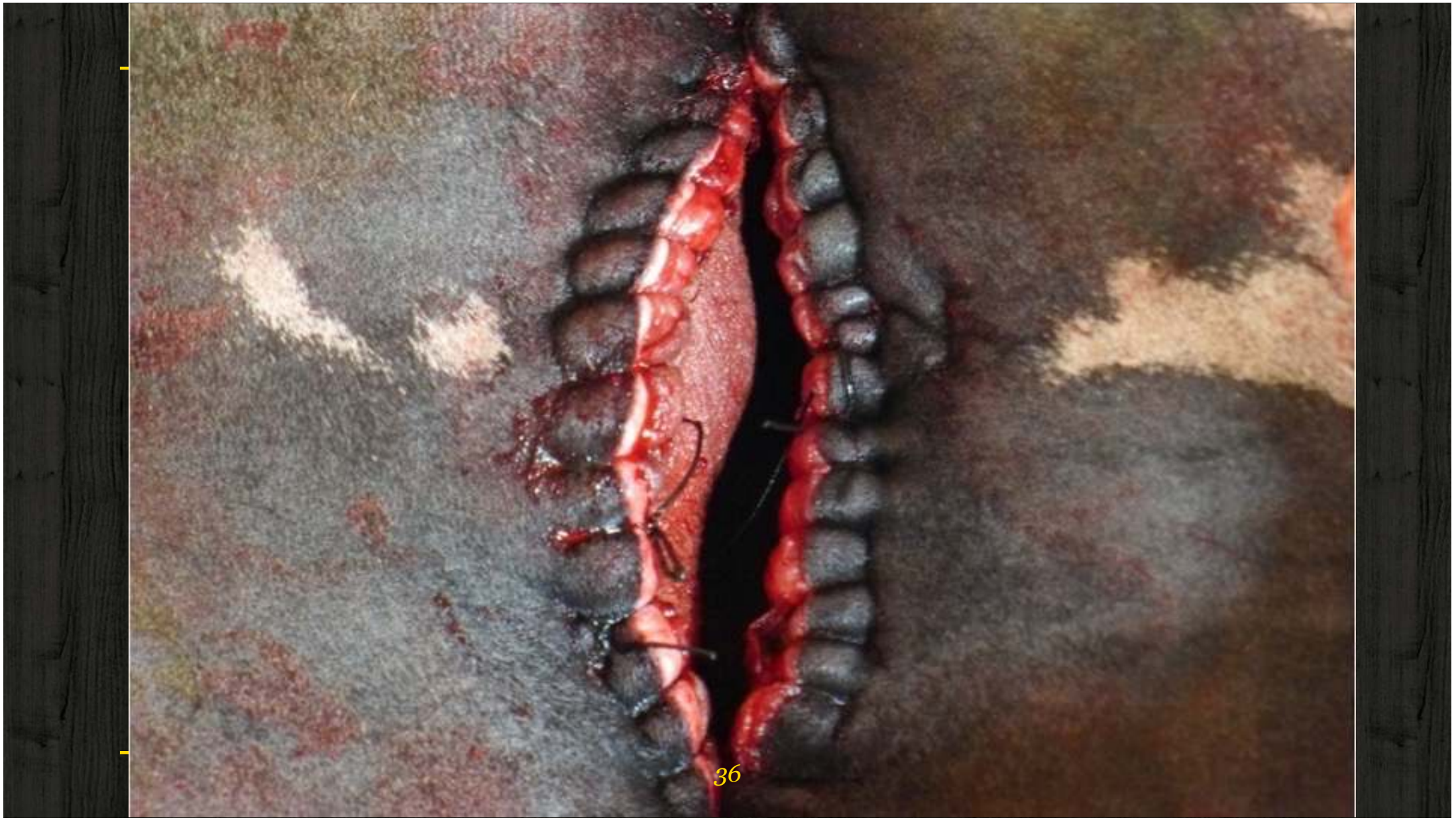




Surgical Technique

4. The rumen is incised with a scalpel.
   - pay attention to bleeding control and preventing rumen contents flowing into the abdominal cavity.
5. The rumen is sutured to the skin using a simple continuous suture method with No. 3 non-absorbable suture.
   - the previous fixing suture can be removed as needed.





Surgical Technique

4. The wound is applied with iodine.
5. Install the rumen fistula and the cap.









Postoperative Management

- The wound is applied with iodine or antibiotics ointment, twice per day, for 4–5 days.
- Administer prophylactic antibiotics (ex. Shotapen®).



Postoperative Management

- About 14 days after the operation, the fistula is temporarily removed, and the sutures are removed.
- Re-install the fistula tube and cap.
- The wound is applied with iodine or antibiotics ointment, twice per day, for further 4–5 days.
- .


瘤胃瘻管拆線









移除傷口壞死組織




以1%碘液仔細擦拭傷口



裝上瘤胃瘻管



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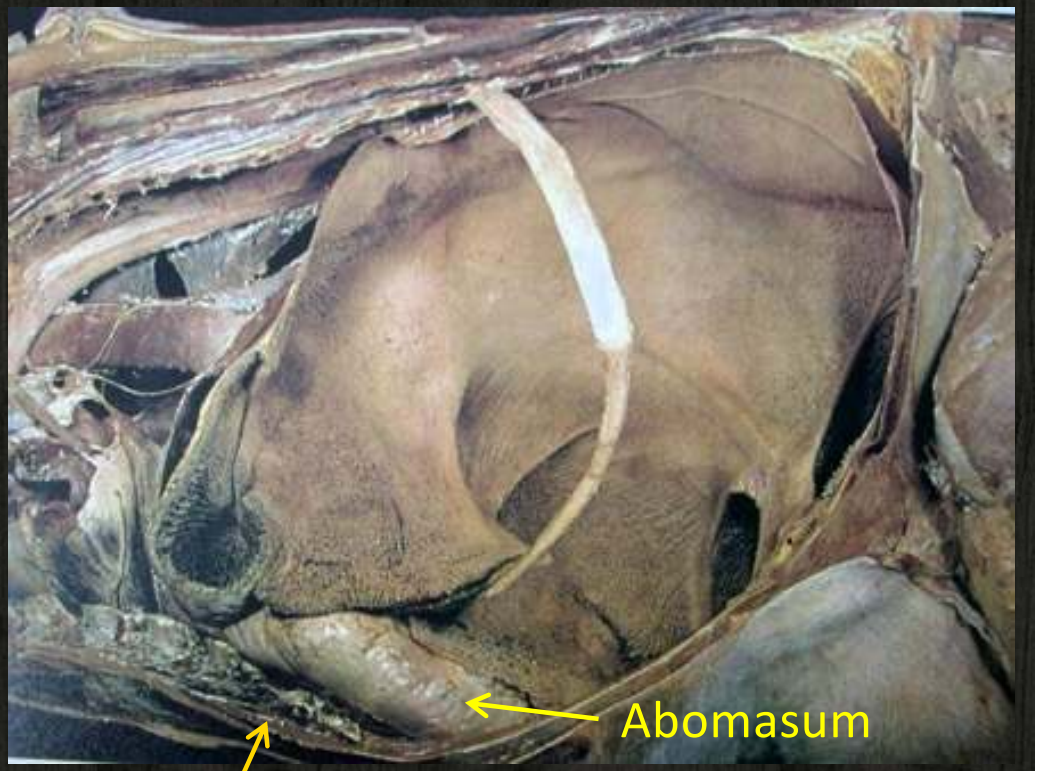
Right-flank Omentopexy

Purpures of Right-Flank Omentopexy

- Left-sided displacement of the abomasum
- Right-sided displacement of the abomasum

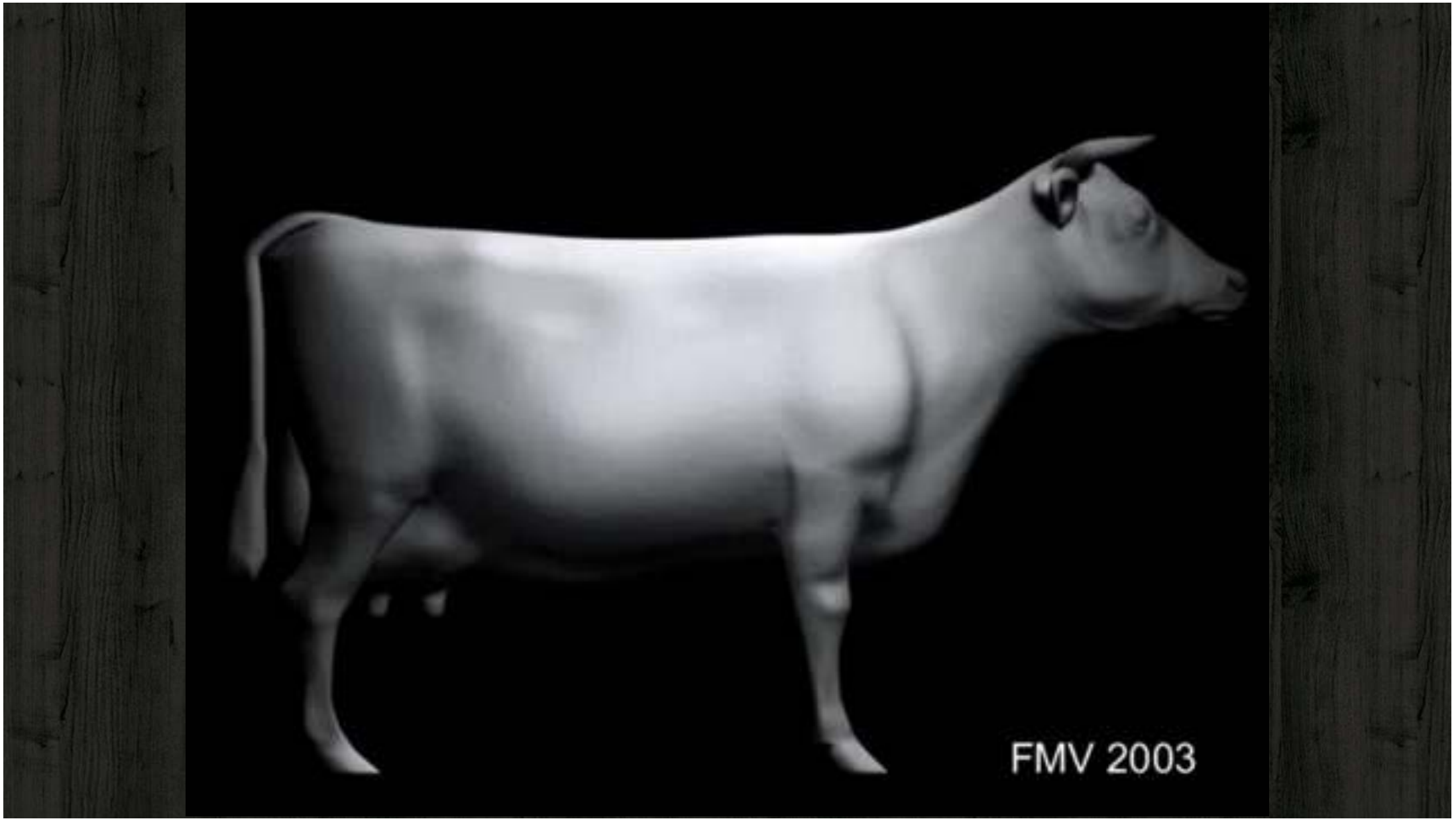
Displacement of the Abomasum

- The fundus of the abomasum is located above the xiphoid process of the sternum, and most of the body of the stomach is located slightly to the right of the midline; the pyloric part is located at the 9th or 10th rib on the right side, extending upward.



Displacement of the Abomasum

- For various reasons, its position shifts to the left and it becomes sandwiched between the left abdominal wall and the rumen, leading to digestive disorders and nutritional imbalances, this condition is called left-sided displacement of abomasum (LDA).
- If it shift to the right side, it is called right-side displacement of abomasum (RDA).


FMV 2003

Aetiology of DA

- Parturition: the most significant contributing factor.
- Atony: precondition.
  - expansion uterus lifts the rumen and also the abomasum, pushing the abomasum forward and located slightly to the left, under the rumen.
  - After delivery, the rumen descends, and if the abomasum is atony, it is pressed down by the rumen and moves to the upper left.

Clinical signs of DA

- Depressed
- Reduced appetite (complete anorexia, reduced consumption of concentrates, or alternating periods of normal appetite and anorexia)
- Feces are often normal to softer (pasty or muddy) than normal but reduced in volume.
- Milk production is reduced rapidly.
- Gradually became thin due to long-term digestive problems.

Diagnosis of DA

- Simultaneous auscultation and percussion reveal a high-tone ping sound over the gas-filled portion of the abomasum.
  - With LDA the area of ping may be anywhere from the lower third of the abdomen in the eighth intercostal space to the paralumbar fossa



Surgical treatment of DA

- Right-flank omentopexy
  (右脅腹大網膜固定術)
- Toggle-pin abomasopexy, blind tack abomasopexy
  (套鎖針第四胃固定術)
- Laparoscopic abomasopexy
  (腹腔鏡第四胃固定術)
- Ventral paramedian abomasopexy
  (腹旁中線第四胃固定術)

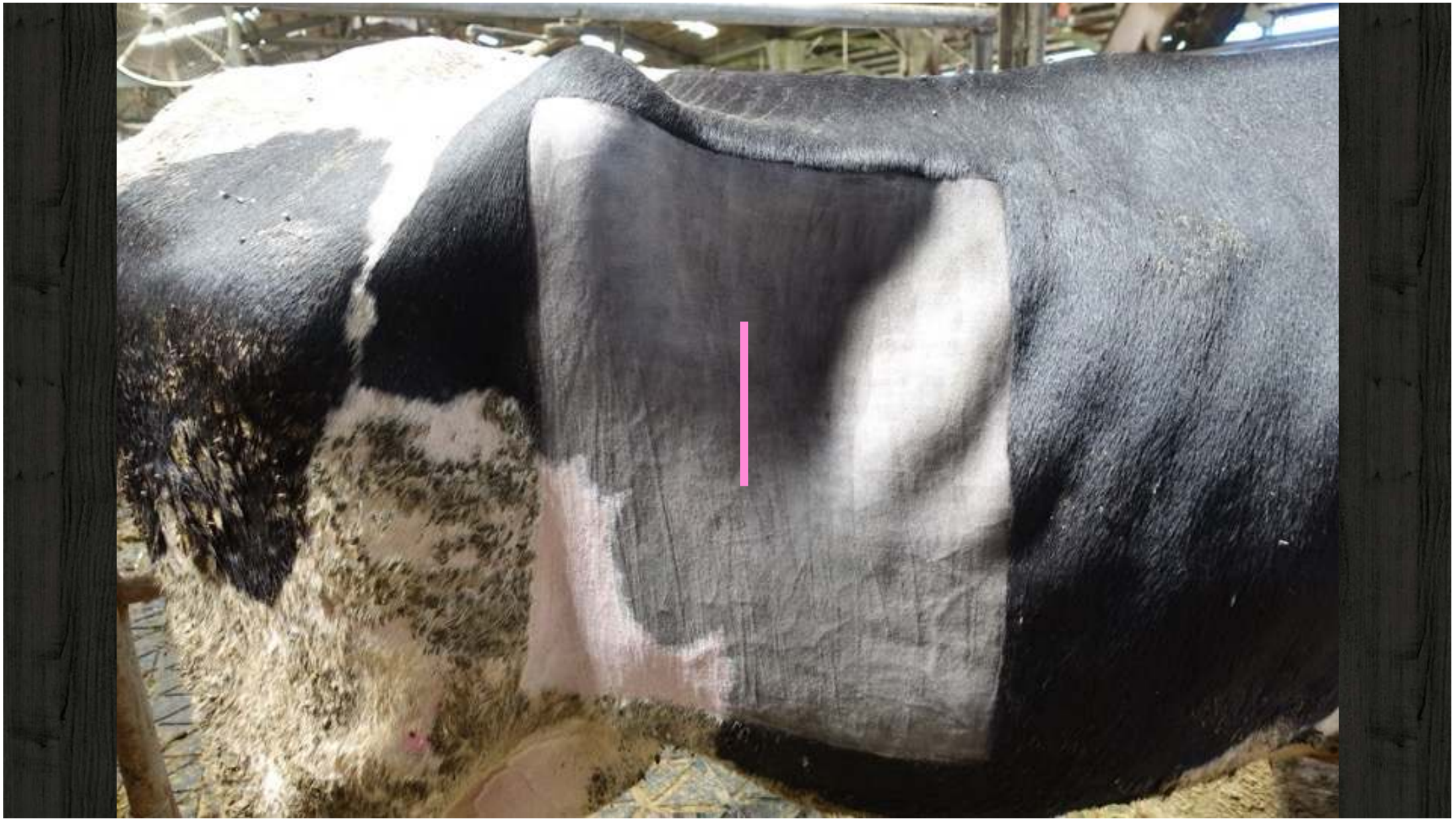
Right-flank omentopexy

Used for almost all cases:

- The cow being able to stand during surgery, and the procedure not being directly sutured to the abomasum
- Fewer labors required
- Low recurrence rate
- Disadvantages: inability to directly visualize the abomasum, the potential for peritonitis, or edema and infection at the surgical site, unable to prevent recurring of RDA .

Right-flank omentopexy procedures

1. Surgical preparation and linear infiltration anesthesia at the right flank.
2. Make a vertical incision about 15 cm long on the right-side flank area.
   - 10 cm behind the last rib and about 20 cm below the transverse process of the lumbar vertebra.

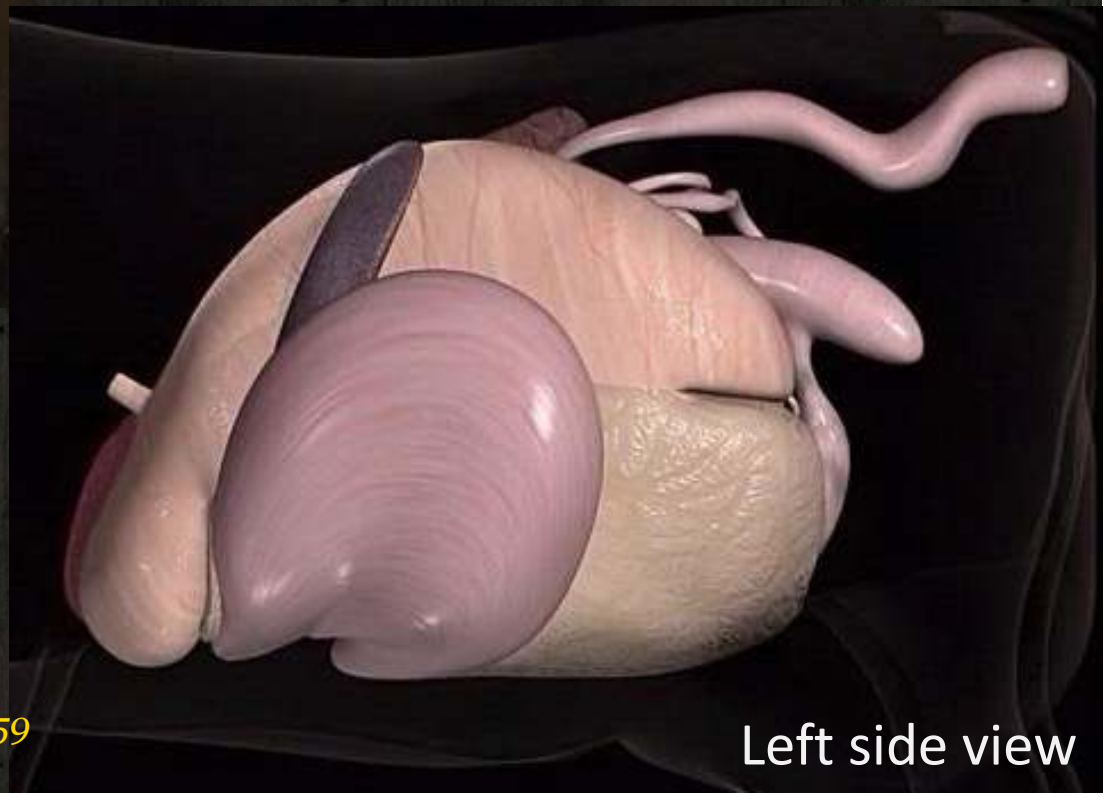


Right-flank omentopexy procedures

3. The external oblique is incised vertically, the internal oblique is bluntly dissected along its direction, and the transverse abdominis and peritoneum are incised together.
4. The left hand is inserted into the abdominal cavity, though the top of the rumen, reaching to the left side of the rumen and the body wall, to palpate and understand the degree of distension of the abomasum and its related location.



- Reaching to the left side of the rumen and the body wall, to recognize the degree of distension of the abomasum and its related location

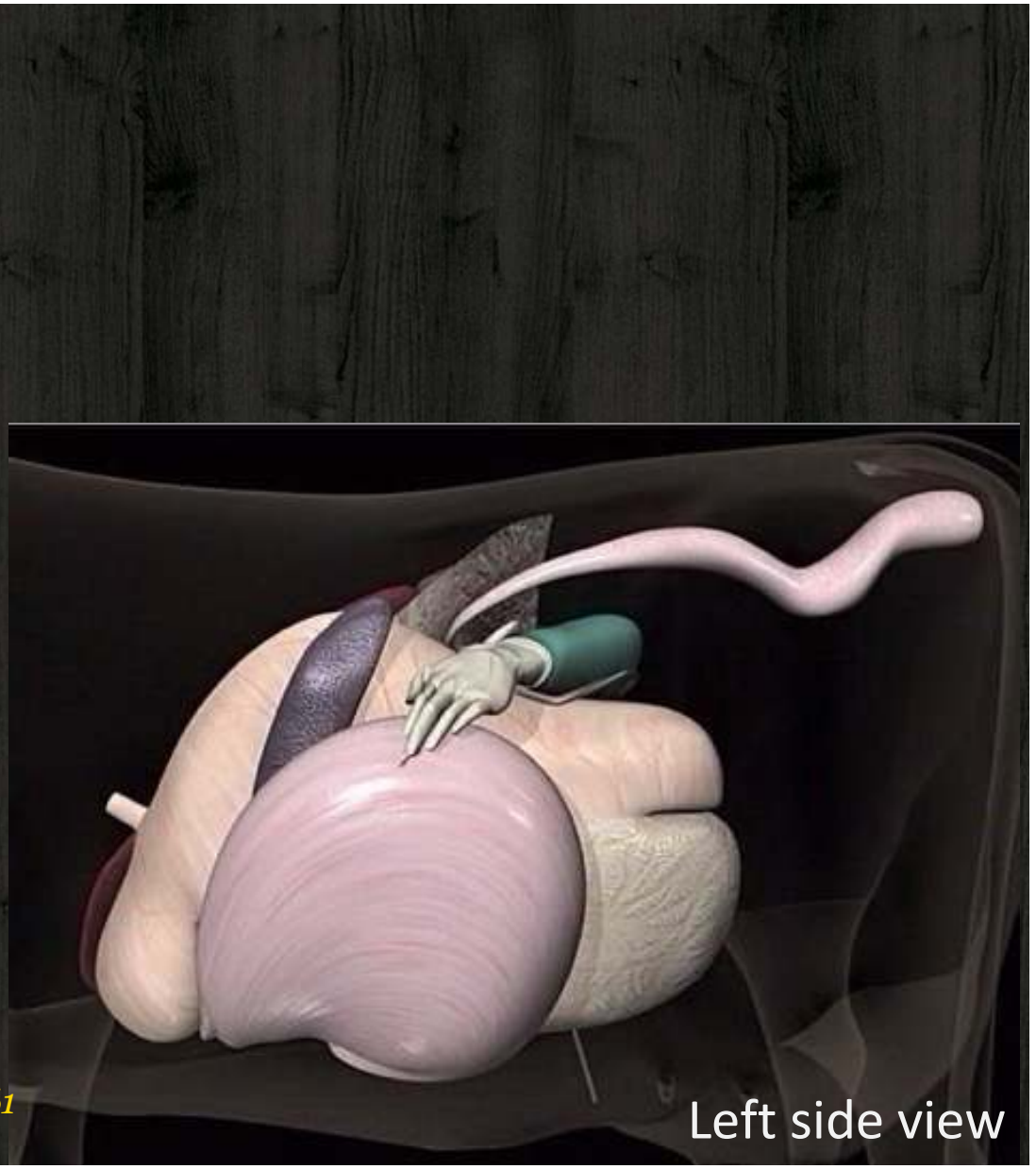


Left side view

Right-flank omentopexy procedures

5. The abomasum may be deflated using a needle with a length of sterile tube attached.
   - The needle is inserted at the most dorsal part of the abomasum.
   - Pressure is applied firmly with hand or the tube may be attached to a suction device, releasing the accumulated gas in the abomasum.
   - Expel as much gas as possible from the abomasum.

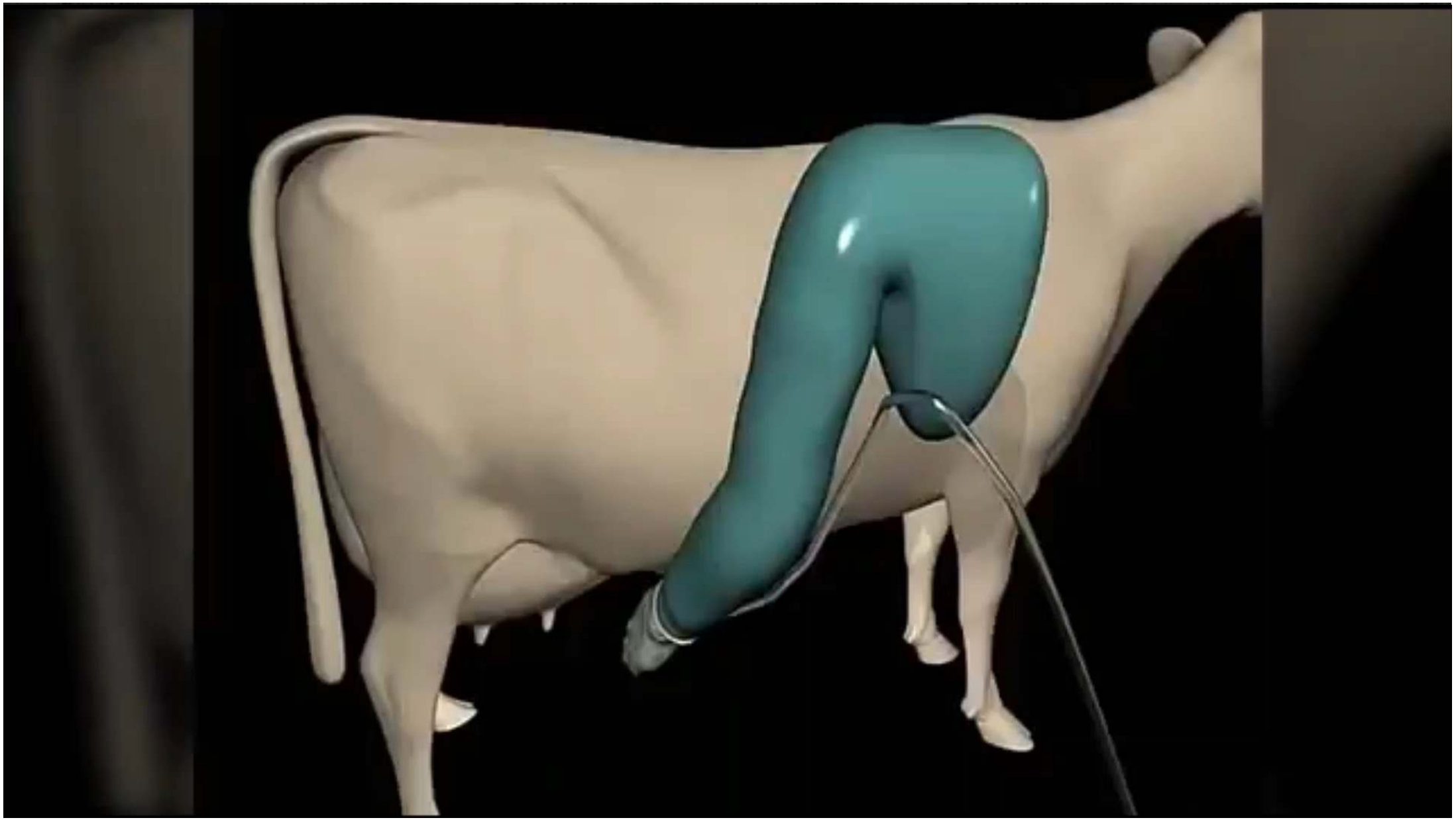




Left side view

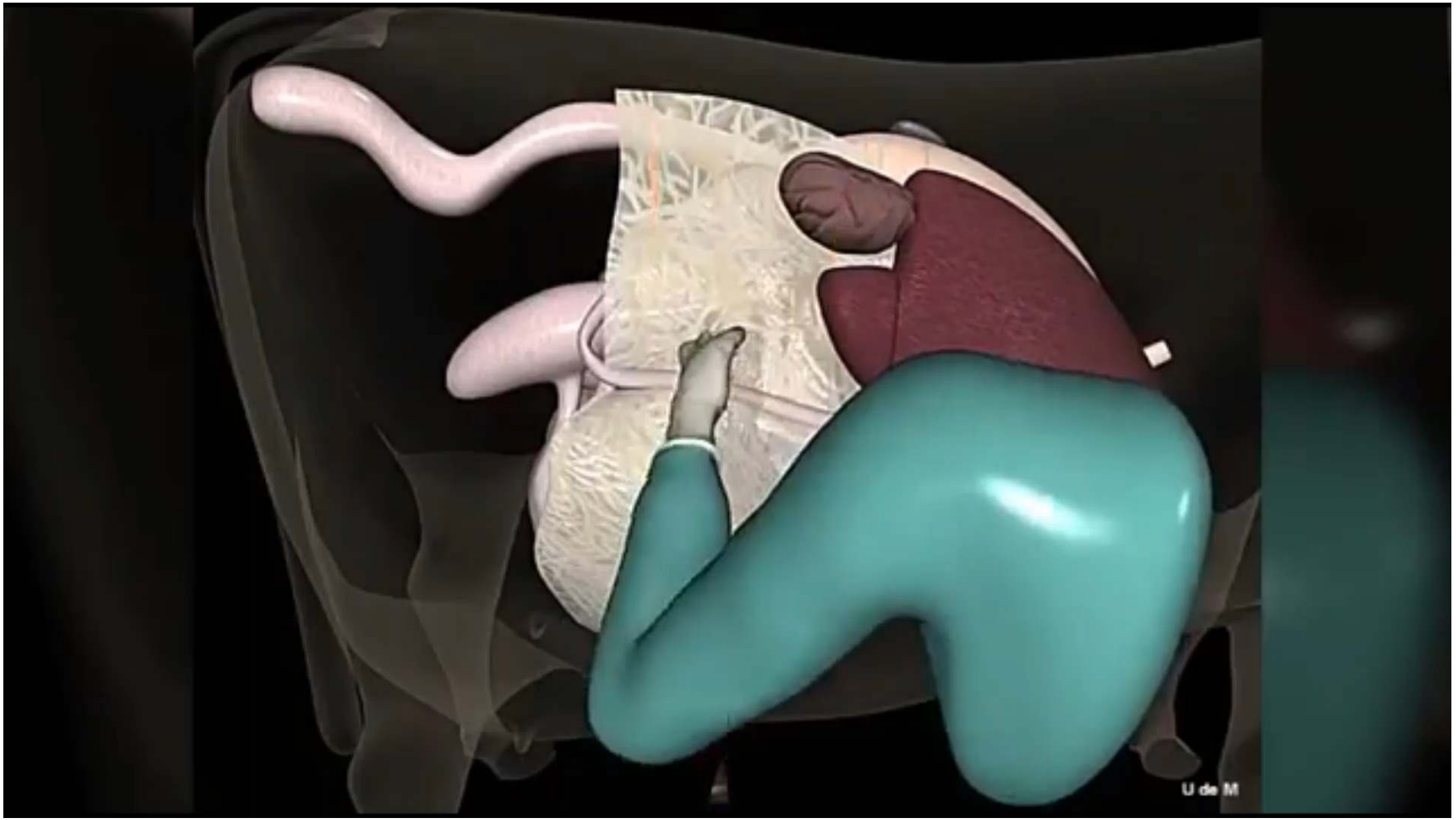






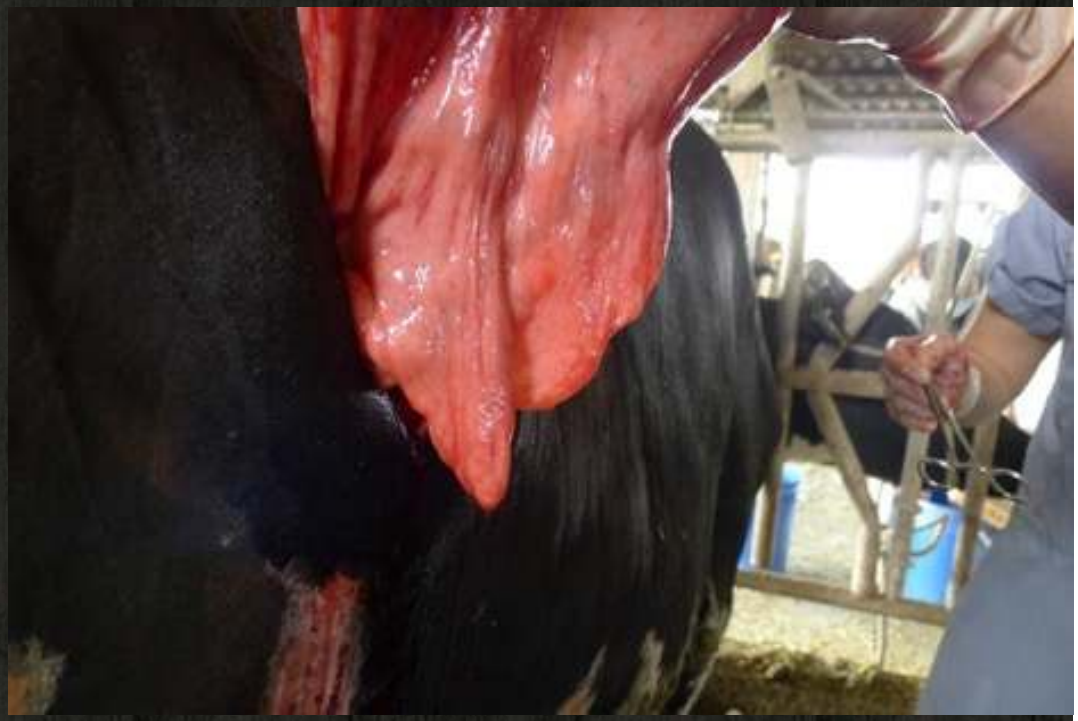
Right-flank omentopexy procedures

6. The needle is withdrawn and is carried back carefully, with the tubing folded to avoid contamination.
7. The hand move to the place between the rumen and the body wall, following the peritoneal surfaces vertically.
8. With the fingers closed, sweep the abomasum back its normal position.
   - Gentle dorsocranial pulling on the omentum is also helpful.
   - Elevate the rumen with the surgeon's elbow to alow abomasum to be pulled along under the rumen.


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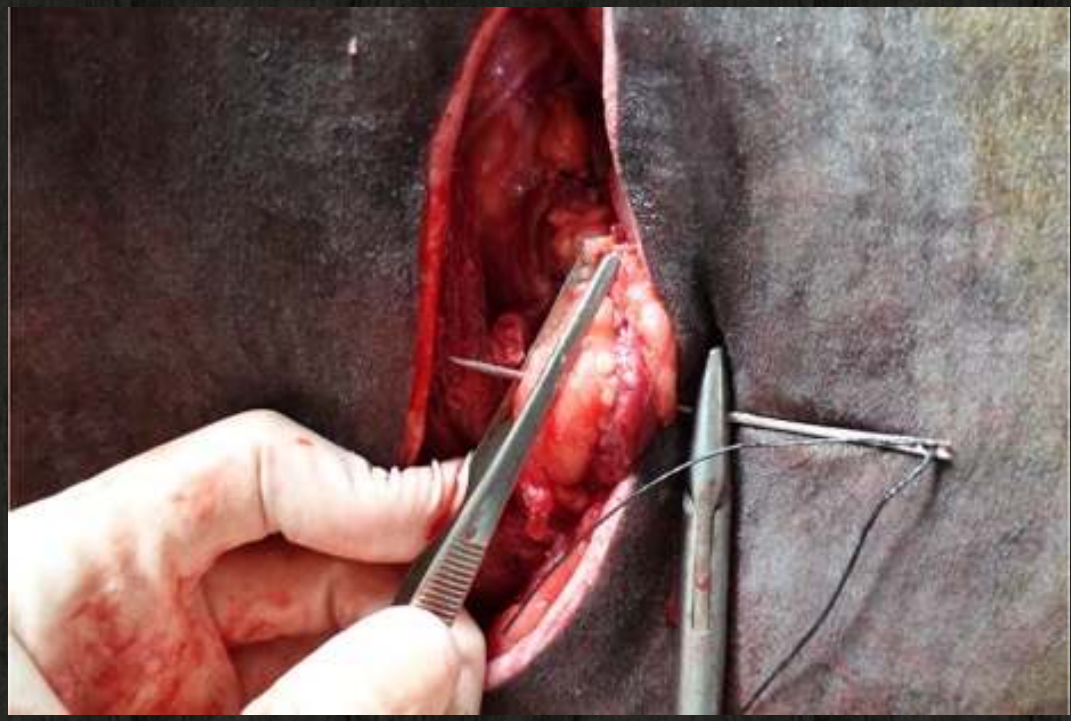
Right-flank omentopexy procedures

9. The omentum is grasped and pulled out through the incision to find the thickened fold of omentum, which is near the duodenum.



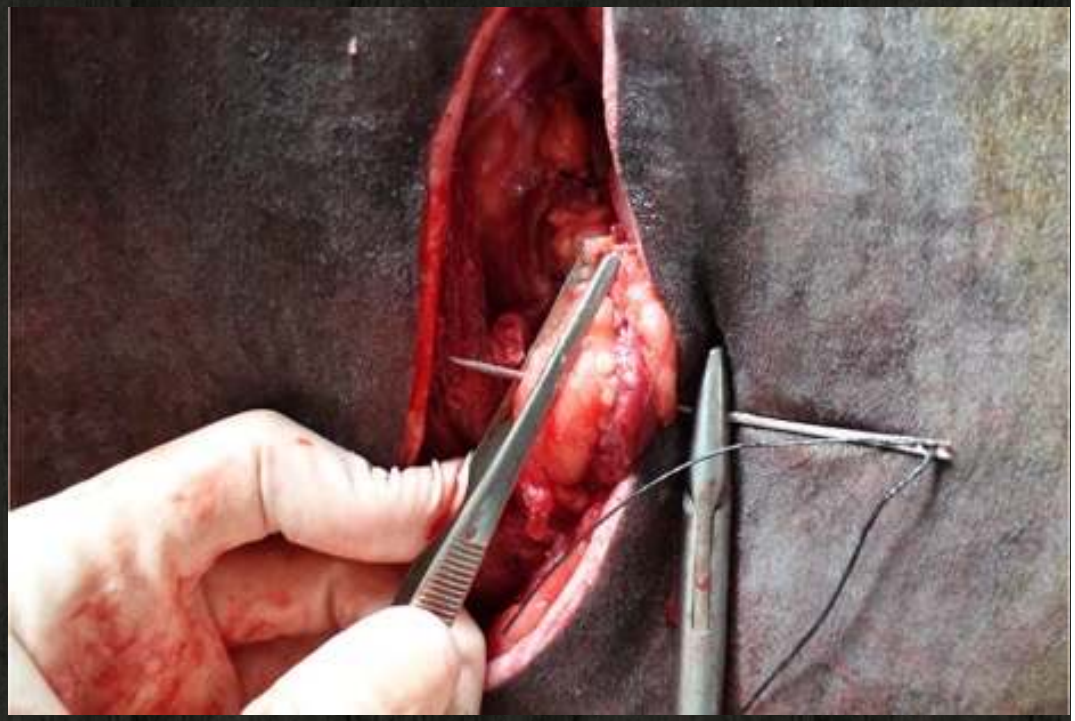
Right-flank omentopexy procedures

10. The fold of omentum is sutured with the transverse abdominis and the peritoneum together using a simple interrupted suture pattern with no. 3 non-absorbable suture.



Right-flank omentopexy procedures

10. The fold of omentum is sutured with the transverse abdominis and the peritoneum together using a simple interrupted suture pattern with no. 3 non-absorbable suture.
11. Closure of the laparotomy incision.



Right-flank omentopexy procedures

10. The fold of omentum is sutured with the transverse abdominis and the peritoneum together using a simple interrupted suture pattern with no. 3 non-absorbable suture.
11. Closure of the laparotomy incision.
12. Postoperative management was followed as previous described.

Toggle-pin abomasopexy

- Economic considerations and quick results are required
- The cow is restrained and laid on her backs, and the location of the distended abomasum is determined by auscultation and percussion.

Toggle-pin abomasopexy

- The advantage of the toggle-pin method is that it allows confirmation of the abomasum by smelling a distinctive odor or measuring the pH of the gastric juice through the cannula.
- Simple LDA can increase appetite and milk production within 48-72 hours using this method.

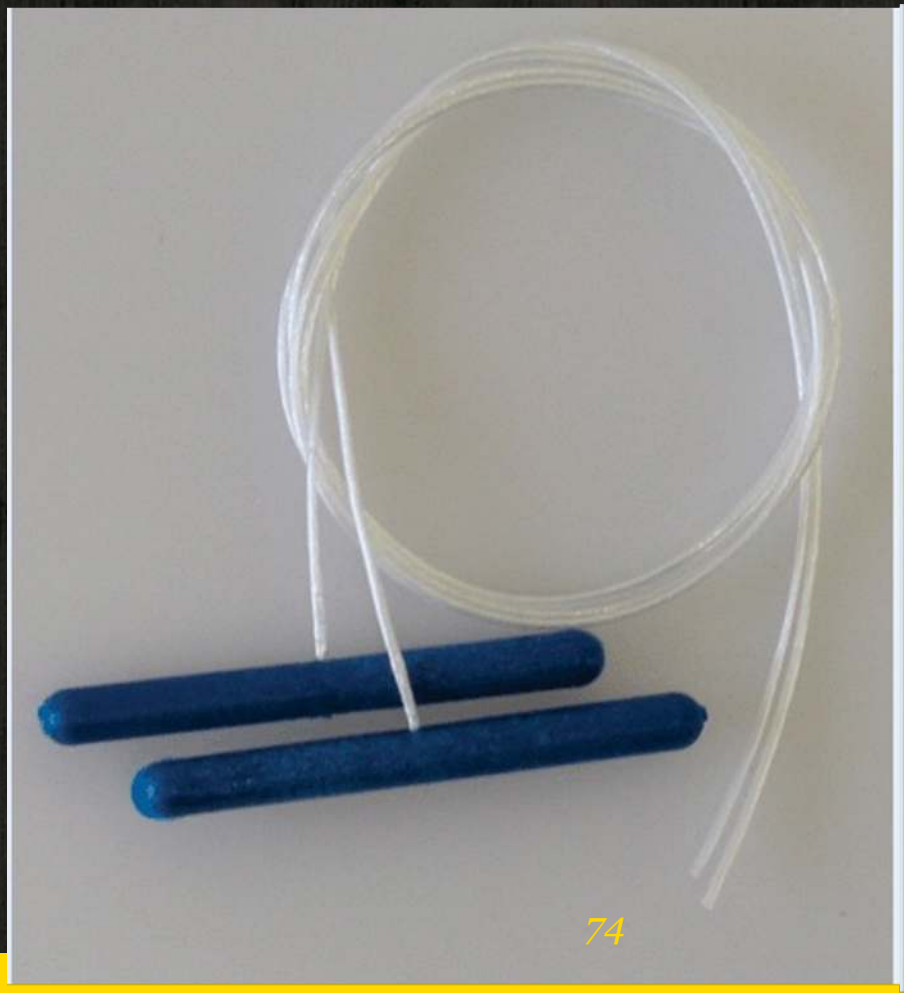
Toggle-pin abomasopexy



Toggle-pin abomasopexy

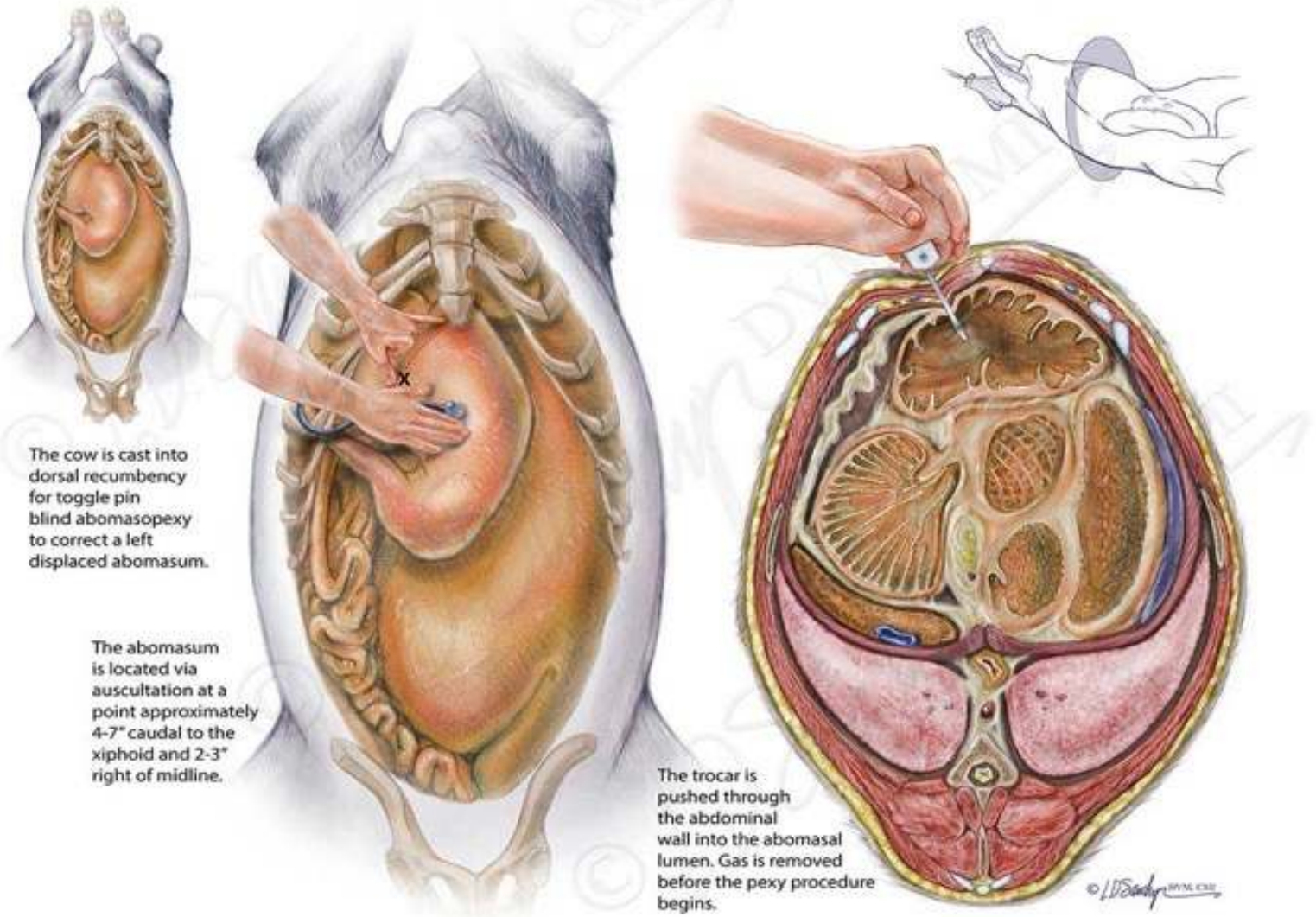


Toggle-pin abomasopexy





https://www.sawchynmi.com/sawchynportfolios/toggle-pin-stabilization-of-bovine-left-displaced-abomasum/







Toggle-pin abomasopexy

- If the sick cow does not show significant improvement in appetite after undergoing toggle-pin abomasopexy, it is best to remove the sutures within 2 days and treat it with other surgical procedures.

Toggle-pin abomasopexy

Disadvantages include:

- Missing the abomasum and instead surrounding the pylorus or the body of the stomach
- The abomasum being fixed in the wrong position
- Fluid flowing from the abomasum into the abdominal cavity
- Puncturing the wrong organ
- Damaging veins in the breast and causing phlebitis
- Peritonitis by contaminants from exogenous or endogenous sources.

Laparoscopic abomasopexy

Advantages:

- Less invasive
- Good success rate
- Low complications
- Shorter operative time
- Fast postoperative recovery

Disadvantages:

- Inability to use in cows with adhesions to the abosamum

Ventral paramedian abomasopexy

- Assistants are needed to hold the cow in place.
- This applies to cattle with ulcers in the fourth stomach that are adhered to the abdominal wall.



The xyphoid, milk veins, and midline are outlined in black.

https://www.researchgate.net/publication/5379234_Surgery_of_the_Abomasum/figures

Ventral paramedian abomasopexy

- Disadvantages:
  - the cow lying on its back
  - the risk of hernia or fistula formation at the incision site
  - contamination of the incision site
  - edema on the lower flank during late pregnancy can also affect the surgery
  - damage to blood vessels in the abdominal lining can impair mammary circulation.

Ventral paramedian abomasopexy

- Contraindications:
  - acute or chronic bronchopneumonia
  - definite musculoskeletal injuries
  - cows in late pregnancy.



Thank you!